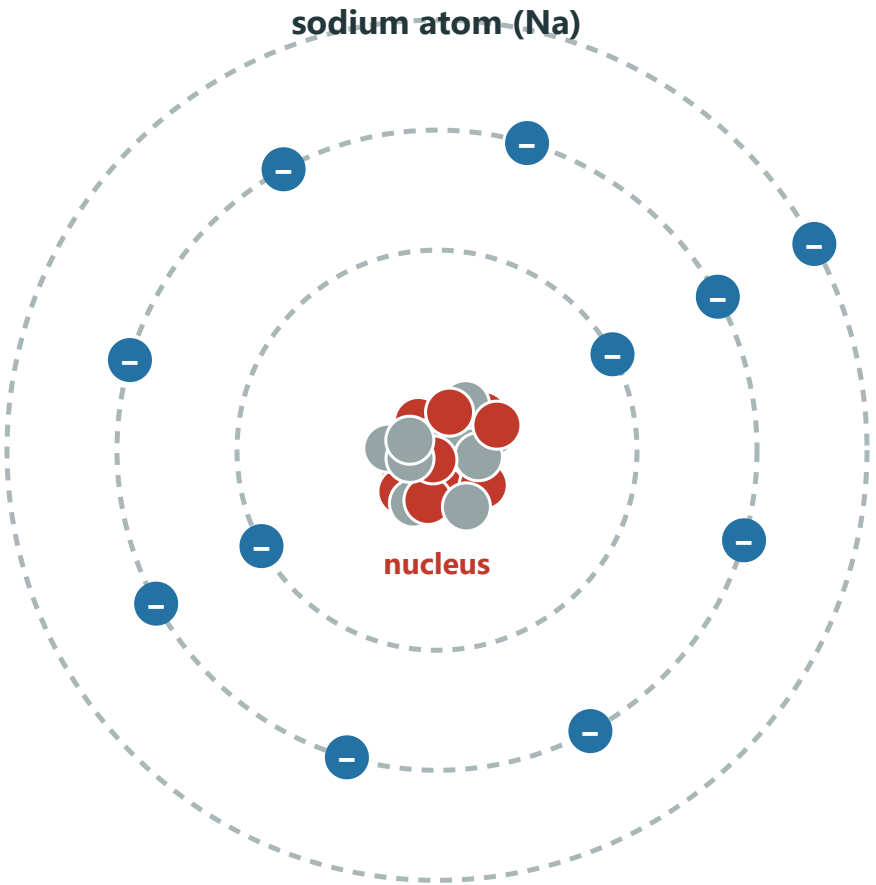
Inside the atom



Particle	Charge	Mass	Where?
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	≈ 0	shells

$^{23}_{11}\mathrm{Na}$

Z = atomic number = number of protons

A = mass number = protons + neutrons

N = A − Z = neutrons → 23 − 11 = **12**

Neutral atom: electrons = protons = **11**